

2024 Formula SAE-A Technical Inspection Sheet

MECHANICAL INSPECTION (ALL CLASSES)

Initial Presentation Scrutineer Name:		Initial Presentation Lane Number:	
1st Reinspection Scrutineer Name:		1st Reinspection Lane Number:	
2nd Reinspection Scrutineer Name:		2nd Reinspection Lane Number:	
3rd Reinspection Scrutineer Name:		3rd Reinspection Lane Number:	

PERSONNEL			Inspection #			
Teams are required to have an Electrical Safety Officer.			Initial ins.	1st reins.	2nd reins.	3rd reins.
MI.1.1	Only team members involved in technical inspection can be in the technical inspection bay	Ask for non-essential people to leave the area				

INERT VEHICLE (EV & AV CLASSES ONLY)			Inspection #			
The Vehicle must be in a condition which prevents unexpected energization of the HV electrical systems or movement of the tractive system For ICE Class vehicles, mark as "N/A"			Initial ins.	1st reins.	2nd reins.	3rd reins.
MI.1.2	Disable the HV Systems - High Voltage Disconnect (HVD)	Check the HVD is removed				
MI.1.3	Disable the HV Systems - Tractive System Master Switch (TSMS) Lock	Check the TSMS Lock is fitted and effective				
MI.1.4	(AV and Dual Purpose only) Disable the ASMS - Autonomous System Master Switch (ASMS) Lock	Check the ASMS Lock is fitted and effective				
DO NOT PROCEED WITH TECHNICAL INSPECTION IF HVD IS INSTALLED OR THE TSMS LOCK IS MISSING						

INSTRUCTIONS FOR SCRUTINEERS						
This section applies to all vehicle classes. If an item is acceptable, initial the "Passed" column. If an item is unacceptable, initial the "Reinspect" column and inform the team members they have not meet the requirements for that item. Teams are allowed to take their vehicle away from technical inspection and address any issues before representing for reinspection. Scrutineers may continue with the inspection when unacceptable items are found, provided it is safe to do so. Teams are required to pass all items before the vehicle can proceed to the Brake Test or compete in Dynamic Events. Scrutineers reserve the right to refuse to proceed with Technical Inspection if a sufficiently serious issue which might jeopardize the safety of the technical inspection team, event personal or any other persons is found at any time.						

MECHANICAL INSPECTION

PRE-REGISTRATION				Reinspect	Passed
MI.2.1	Team completed inspection records	Teams must have self-completed the Technical Inspection Checklist and present a hardcopy	Ask for hardcopy of TI inspection sheet		
MI.2.2		Teams must provide evidence of egress times achieved by all drivers.	Ask for hardcopy record of egress times		
MI.2.3	Impact Attenuator	Teams must present a sample of their tested Impact Attenuator, or detailed photographs. (Including Standard IA if required due to Bulkhead configuration.)	Ask to see the sample impact attenuator or photos.		
TECHNICAL INSPECTION STICKER AND COMPETITION SPONSOR STICKERS				Reinspect	Passed
MI.3.1	Technical Inspection Vehicle Panel	The Technical Inspection Sticker must be installed on the top surface of the nose of the vehicle on the vehicle centerline. The Technical Inspection Sticker must be used as provided and not modified. The Technical Inspection Sticker must be clearly visible.	Visual Check		
MI.3.2	Competition Sponsor Stickers	The Competition Sponsor Sticker must be installed on the top surface of the nose of the vehicle on the vehicle centerline. The Competition Sponsor Sticker must be used as provided and not modified. The Competition Sponsor Sticker must be clearly visible.	Visual Check		

MECHANICAL INSPECTION

WHEELS & TYRES

Teams are required to bring a sample of each dry and rain tyre for use. Teams are allowed ONE SET of each type of tyre.

Teams may have different tire sizes front and rear, but left and right tires must be identical.

Teams may replace damaged tires but need to re-present to TI and demonstrate they are identical to the make size and compound listed below.

				Reinspect	Passed
MI.4.1	Dry Tyres	Inspector to record tyre make, size and compound. Make: Size: Compound: Apply tamper proof sticker to tire and wheel rim	Visual Check		
MI.4.2	Rain Tyres	Inspector to record tyre make, size and compound. Minimum tread depth of 2.4mm. Make: Size: Compound: Apply tamper proof sticker to tire and wheel rim	Visual Check		
MI.4.3	Wheels	Four wheels not in a line. 203.2 mm (8 inch) minimum diameter. A single wheel nut must have positive retainer. Check correct wheel nuts, tapered nuts, washers and spacers.	Visual Check		

MECHANICAL INSPECTION

DRIVER'S EQUIPMENT				Reinspect	Passed
Driver's equipment must be in good usable condition and conforming to the standards listed below					
MI.5.1	Helmets	Must comply with Motorsport Australia, Technical Appendix, Schedule D, 2.1 Helmet standards A and B as listed	Visual Check		
MI.5.2	Frontal Head Restraint	Not compulsory - FIA 8858-2010, 8858-2002	Visual Check		
MI.5.3	Suits	Single piece - FIA 8856-2018, FIA 8856-2000, FIA 1986, SFi 3.2A/5 or superior SFi standard.	Visual Check		
MI.5.4	Underwear	Not compulsory - recommended. FIA 8856-2000, 8856-2018	Visual Check		
MI.5.5	Balaclava - hair cover	FIA 8856-2000, FIA 8856-2018	Visual Check		
MI.5.6	Socks	FIA 8856-2000, FIA 8856-2018	Visual Check		
MI.5.7	Shoes	FIA 8856-2000, FIA 8856-2018, SFi 3.3 or shoes with leather uppers.	Visual Check		
MI.5.8	Gloves	FIA 8856-2000, FIA 8856-2018 or leather.	Visual Check		
MI.5.9	Arm Restraints	SFi 3.3	Visual Check		
DRIVER'S COMMUNICATION				Reinspect	Passed
MI.6.1	Interconnected to vehicle	If part of driver's equipment, then it must not be attached to driver's helmet unless fitted by helmet OEM.	Visual Check		
MI.6.2	Telemetry	If fitted, may only be used to monitor vehicle state.	Visual Check		

MECHANICAL INSPECTION

EXTERIOR GENERAL - VEHICLE ID & EQUIPMENT				Reinspect	Passed
MI.7.1	Fire extinguishers	Two hand-held, 0.9kg (2lb) minimum, dry powder, dry chemical, Must see both. One extinguisher mounted on Push Bar. Portable fire extinguishers must not be mounted to the vehicle. Dedicated On-board extinguishing systems encouraged	Visual Check		
MI.7.2	Push bar	With car, detachable, push and pull for 2 people standing behind car.	Visual Check		
MI.7.3	Jacking Point	Must have an exposed tube 30cm long, 2.5-2.9cm OD. Painted Orange. Rear wheels must come off the ground using Quick-jack (200mm lift).	Visual Check		
MI.7.4	Body and Styling	Open wheeled, open cockpit, formula styled body. Vertical keep out zones 75mm in front & behind tyres, no aero exceptions, tyres unobstructed from sides.	Visual Check		
MI.7.5	Car Numbers	Numbers must appear on front (upper surface of the nose, on centerline) and both sides. Not obscured. Min height 150mm with 25mm spacing from edges. Helvetica Bold. In Dayglo Yellow. Black background - Round, Oval, Square or Rectangle.	Visual Check		
MI.7.6	School Name and Other Decals	School name or initials, min height 50mm on both sides in Roman letters.	Visual Check		
EXTERIOR GENERAL - BODYWORK & AERODYNAMICS				Reinspect	Passed
MI.8.1	Wings	Securely mounted, should not wiggle when gently touched, especially side to side. If in question inform chief. Affix inspection sticker/s to front and /or rear wings/ aerodynamic devices.	Visual Check		
MI.8.2	Wing Edges	Horizontal leading edges min 5mm radius, vertical forward facing edges min 3mm radius. Radius must be part of structure or permanently attached.	Physical check with radius gauge		
MI.8.3	Bodywork	Minimum 38mm radius on nose, No large openings in bodywork entering into driver compartment in front of or alongside driver. 75mm clearance Front and Back of tyres viewed from above.	Physical check with radius gauge		
MI.8.4	Wheel base	Minimum 1525mm (60 inch)	Physical check with tape measure		
MI.8.5	Aerodynamics	All Aero devices, wings, undertrays, splitters, max 70cm forward of front tyres, max 250mm rearward of rear tyres. Front wings no wider than outside of front tyres. Rear wings no wider than inside of rear tyres. Under trays no wider than line between front and rear tyres. No power ground effects.	Physical Check with Gauges		
MI.8.6	Aero Vertical Height	Rear wing max 1200mm above ground (inc end plates). Front wing max 250mm above ground. No bodywork or aero higher than 500mm between axles(except centre 800mm of car, ie, cockpit panels)	Physical Check with Gauges		
MI.8.7	Cameras	If >0.25kg, must be secured by two points. No cameras mounted on helmets.	Visual Check		

MECHANICAL INSPECTION

EXTERIOR GENERAL - CHASSIS & STRUCTURAL				Reinspect	Passed
MI.9.1	Alternative frame	If alternative tube size/material, approved Structural Equivalency Sheet (SES) required. If using Alternative Frame Rules, no titanium or magnesium in primary structure.	Visual / Physical Check		
MI.9.2	Inspection holes	Inspector may use ultrasound to measure wall thickness and/or ask 4.5mm hole be drilled.	Visual / Physical Check		
MI.9.3	Main hoop	MUST BE STEEL. 1.00" OD x 0.095" wall or 25mm OD x 2.5mm wall. Must be 1 piece & extend to lowest frame member. 380mm apart (inside dim.) where attaches to bottom tubes of the Major Structure. Above Major Structure, must be within 10° of vertical. No part angled rearwards more than 10° from vertical. Smooth bends with no wrinkles. Attachment to Monocoque/Composite must be mechanically affixed.	Visual / Physical Check		
MI.9.4	Main hoop bracing	MUST BE STEEL. One brace each side, 1.00" x 0.065" or 25mm x 1.6mm minimum, attached within 16cm of top. Minimum 30 deg. included angle with hoop. If main hoop is not vertical, bracing must not be on same side of vertical as main hoop. No bends. No rod-ends. Proper construction for removable braces (capping etc.) on BOTH ENDS. Must take load back to bottom of main hoop and node of upper side-impact tube thru proper triangulated structure. If any item which is outside the envelope of the Primary structure is attached to the Main Hoop braces not at a node, additional bracing must be added to prevent bending loads in the braces in any rollover attitude. (eg.. suspension mounts, radiators or wings).	Visual / Physical Check		
MI.9.5	Bolted joints	Edge of any bolt hole located > 1.5 x hole diameter from nearest edge of the material (Primary structure joints only)	Visual / Physical Check		
MI.9.6	Shoulder harness mounting bar	1.00" OD x 0.095" wall or 25mm OD x 2.5mm wall steel or equiv. Gussets or braces if not straight to main hoop	Visual / Physical Check		
MI.9.7	Front hoop	Must be closed section metal tube. 1.00" OD x 0.095" wall or 25mm OD x 2.5mm wall steel, or equiv. Can be multi-piece. Must extend down to lowest frame member. Maximum 20 deg. to vertical. No lower than top of steering wheel. Maximum 250mm horizontal distance to steering wheel.	Visual / Physical Check		
MI.9.8	Front hoop bracing	Two forward facing braces, 1.00" OD x 0.065" or 25mm OD x 1.6mm steel or equivalent, attached within 5cm of top. Extra rearward bracing required if Front Hoop leans backwards more than 10 degrees.	Visual / Physical Check		

MECHANICAL INSPECTION

EXTERIOR GENERAL - CHASSIS & STRUCTURAL CONTINUED...				Reinspect	Passed
MI.9.9	Other side tubes	Design prevents driver's neck hitting bracing or other side tubes.	Visual Check		
MI.9.10	Side impact protection	Comprised of 3 or more tube members on each side. Must connect to main and front hoops. Lower tube to bottom of both hoops. Lower tube can be lower frame member. Upper tube must be in a zone parallel to ground between 240mm and 320mm above the lowest point of the top surface of the Lower side impact member. At least one diagonal per side must connect the upper and lower members between the main and front hoops. All tubes to be 1.0" OD x 0.065" wall or 25mm OD x 1.6mm wall steel or equivalent. If Upper Side Impact tube is multi piece or bent, must have triangulating supporting tube from the furthest part of deviation from the straight line, back to a node on the chassis and the bent tube and support must be minimum 35mm x 1.2mm;	Visual / Physical Check with gauge.		
MI.9.11	Front bulkhead	1.0" OD x 0.065" wall, or 25mm x 1.6mm wall, steel tube or equiv. No non-crushable objects forward of bulkhead.	Visual Check		
MI.9.12	Front bulkhead support	Support back to front roll hoop; minimum 3 tubes per side, all 1.00" OD x 0.049" wall steel tube or equivalent 1 bottom; 1 top within 50mm of top of bulkhead, and connecting within 4" above and 2" below upper side impact support (SIS) tube; 1 or more node-to-node diagonal to completely triangulate connections to upper and lower side impact support (SIS) tubes. (25mm x 1.5mm and 26mm x 1.2mm metric tubes OK)	Visual / Physical Check		
MI.9.13	Impact Attenuator	Impact Attenuator forward of bulkhead, 200mm long x 200mm wide x 100mm high. No wing supports through the IA. Bonded, or bolted to Plate with eight 8mm bolts plus additional support.	Visual / Physical Check		
MI.9.14	Impact Attenuator Mounting. Anti intrusion plate	All vehicles must have 1.5mm steel, 4mm Al, or approved equivalent IA anti-intrusion plate. Plate must be capable of taking transverse and vertical loads (welded or minimum eight 8mm bolts). Same size as outside dimensions of Front Bulkhead if bolted or to tube c/l if welded. If the outside profile of the Front Bulkhead is greater than 400 mm x 350 mm; or a Standard Impact Attenuator is used, requires diagonal brace unless test results provided to show Al plate deflection was less than 25 mm. The Standard IA must be attached only by bonding	Visual / Physical Check Team to present test results if required.		
MI.9.15	Monocoque / Composite	Must see laminate test specimens (2 or more) for both side impact support (SIS) and primary structure constructions. Steel backing plates (>2mm thick) used at attachment points.	Visual / Physical Check		
MI.9.16	Thermal Protection	Protection when entering or exiting vehicle, exhaust, coolant hoses or through seat or floor.	Visual Check		

MECHANICAL INSPECTION

VEHICLE CONTROLS - STEERING, SUSPENSION & BRAKES				Reinspect	Passed
MI.10.1	Ground clearance	Sufficient clearance so that no part of the car other than the tyres will contact the track surface.	Visual / Physical Check		
MI.10.2	Suspension	Fully operational with dampers front and rear; 50mm minimum wheel travel with driver in vehicle	Visual / Physical Check		
MI.10.3	Suspension pickup points	Inspected thoroughly for integrity.	Visual / Physical Check		
MI.10.4	Brakes	Dual hydraulic system & reservoirs, operating all 4 wheels, (one brake on limited slip OK). System protected by structure/shields from d/train failure & minor collisions. No plastic brake lines or brake-by-wire. No parts below chassis/tub in side view. Brake pedal (and seat) capable of 2000N (450 lbs-f) with no failures (tested only by organizers.)	Visual / Physical Check		
MI.10.5	Brake Light	One (!) RED brake light, clearly visible from the rear; on vehicles centreline; height between wheel centreline & driver's shoulders. Round, triangle, or rectangular on black background. 15cm ² minimum illuminated area.	Inspect		
MI.10.6	Steering wheel	Continuous perimeter, near round (no concave sections) with driver operable quick disconnect. 25cm max from Front Hoop.	Visual / Physical Check		
MI.10.7	Steering	All steerable wheels must have positive stops to prevent linkage lock-up or tyres contacting any part of the car. 7 degrees maximum free play at the steering wheel. NO STEER-BY-WIRE on front wheels. (AVs may have an actuator on the steering, but the steering wheel must be directly mechanically attached to the wheels.) Rear steer limited to 6° total, with mechanical stops. No bonded joints in column.	Visual / Physical Check		
MI.10.8	Fasteners	Intake manifold, fuel rail, steering, braking, impact attenuator (IA), EV accumulator (battery), harness & suspension system use SAE Grade 5, Metric Grade 8.8 or higher (AN/MS) with visible positive locking mechanisms, no Loctite or lock washers. Minimum of 2 exposed threads. Rod ends in single shear are captured by a washer larger than the ball diameter. Adjustable rod ends have jam nuts to prevent loosening. No button head cap, pan head or round head screws in critical locations, e.g., cage structure or harness mount. Nylon locknuts not for use above 80 °C, i.e., near exhaust.	Visual / Physical Check		

INTERIOR - DRIVER ACCOMMODATION				Reinspect	Passed
MI.11.1	Lap belt mounting	Must pass over pelvic area at between 45-65 deg. to horizontal for upright driver, 60-80 deg. for reclined. Pivoting mounting with eye bolt or shoulder bolt attached securely to Primary Structure.	Visual / Physical Check		
MI.11.2	Driver restraint harness	SFI 16.1, SFI 16.5 or FIA specs 8853/98 or 8853/2016 5, 6 or 7 point and be labelled. 50mm wide shoulder belts OK with FHR. 50mm lap belts OK for FIA & SFI 16.5, All lap belts must have Quick Adjusters. Reclined drivers must have 6 or 7 point, and Quick Adjuster sub-belts or 2 sets of sub belts. Sub belts cannot touch frame tubes or holes in seat. SFI Belts expire 2yr from manufacture date/month; FIA 8853/98 5 years after expiry year marked on label.	Visual / Physical Check		
MI.11.3	Harness mounts	No belts can pass through a firewall. (Belts must mount on driver side of firewalls.) All belts attached securely to primary structure - 1.00" OD x 0.065" steel tube minimum. Any tabs to be 1.0" x 0.063" thick minimum. Double shear preferred. Bolt-on tabs use minimum of two 1/4" diameter Gr 5 bolts.	Visual / Physical Check		

MECHANICAL INSPECTION

INTERIOR - DRIVER ACCOMMODATION CONTINUED...				Reinspect	Passed
MI.11.4	Firewall	Fire resistant material; must separate driver (line-of-sight up to mid-height of driver's helmet) from fuel, cooling & oil systems. Wire/cable pass-throughs OK with grommets. Multiple panels OK w/ gaps sealed. No gaps at sides or bottom	Visual / Physical Check		
MI.11.5	Floor closeout panel	Required from foot area to firewall; solid, non-brittle material; multiple panels are OK if gaps less than 3mm. System Sealing: 1 or 2 drain holes min 25mm must be provided in the structure or belly pan at locations, lowest point of chassis or rearward of driver position forward of any fuel or liquids.	Visual / Physical Check		
MI.11.6	Belt attachment fasteners	Attachment bolt must be a minimum of 10mm Metric Grade 8.8 (3/8" SAE Grade 5). Applies to each belts.	Visual / Physical Check		
MI.11.7	Head restraint / head rest	Meets SFI or FIA listed materials. Min 38mm thick, min 15cm wide, min height 28cm. Max 25mm from helmet, Helmet contact point 50mm minimum from any edge. Adjustable or changeable for different drivers.	Visual / Physical Check		
MI.11.8	Roll bar padding	Meets SFI or FIA listed materials. Rollbar or bracing that could be hit by driver's helmet must be covered with 12mm thick (hard) padding. Pipe insulation and foam not permitted.	Visual / Physical Check		
MI.11.9	Vehicle controls	All controls, including shifter, must be inside cockpit. No hands, arms or elbows outside side impact system to actuate.	Visual / Physical Check		
MI.11.10	Visibility	100 deg. minimum field either side. Head rotation OK or mirrors. If mirrors, must be firmly installed and adjusted	Visual / Physical Check		
MI.11.11	Drivers' foot protection	Feet must be rearward of the Front Bulkhead and no part of shoes or legs above or outside the Major Structure in side or front views when touching pedals.	Visual / Physical Check		
MI.11.12	Drivers' leg protection	Covers inside cockpit over sharp parts or moving suspension and steering components.	Visual / Physical Check		
POWERTRAIN				Reinspect	Passed
MI.12.1	Scatter shield materials	For chains, 2.7mm minimum thick STEEL, 3 x chain width. For belts, 3mm minimum thick aluminum 6061-T6, 1.7 x belt width	Visual / Physical Check		
MI.12.2	Scatter shields general	Required for clutches, chains, belts, CVT rotating parts, etc. No holes. 6mm diameter M8.8 or 1/4" diameter Grade 5 fasteners minimum. End parallel to lowest part of front and rear sprockets	Visual / Physical Check		
MI.12.3	Catch tanks	Coolant overflow, crankcase breather & lube system vents must have separate catch tanks. 1 qt minimum each. 100°C material. Behind firewall, below shoulder level. 3mm minimum diameter vent, away from driver. PCV OK if routed to intake sys upstream of restrictor. Cannot attach breather to exhaust. Trans or diff., unless sealed, require 50 mL catch bottle.	Visual / Physical Check		

MECHANICAL INSPECTION

SPECIALISED TESTS				Reinspect	Passed
MI.13.1	Driver template position PERCY	Seat adjusted to rear most position, pedals to forward most position, Bottom circle placed in seat bottom and between centre of circle and rear most face of the pedals is no less than 915mm. This dimension must be retained. If not it must comply before Dynamic events. Loss of points for Design Score.	Visual / Physical Check		
MI.13.2	Cockpit opening	Template passes down from above cockpit to 25mm below lowest point of the top of the Side Impact Structure to less than or equal to 320mm above the lowest point inside the cockpit.. Steering wheel & column, seat & padding may be removed. No removing firewall. Fore/aft translation of template OK.	Visual / Physical Check		
MI.13.3	Cockpit Internal cross section	Pedals in most forward position. Vertically and may be moved horizontally, Template to pass from cockpit to 100mm rear of pedals. Steering wheel and padding removable with no tools & with driver in seat. Seat remains, Steering wheel can be removed.	Visual / Physical Check		

MECHANICAL INSPECTION

NOTES

INSTRUCTIONS FOR SCRUTINEERS - END OF MECHANICAL INSPECTION (ALL CLASSES)

If the team have passed all items in this section, go to the second page of this document and initial the relevant box, noting the date and time the team passed this section.

- EV and dual purpose AV Classes, proceed with Driver Egress.
- For ICE Class, proceed with ICE Class Inspection.

If the team have items requiring reinspection (did not pass all items), go to the second page of this document and initial the relevant box noting the time and date the team was released from the inspection. The team will need to rectify the reinspection items and may return for reinspection when they are ready.

Teams may proceed to ICE Class Inspection or Driver Egress if it is safe for them to do so.

Reinspection is completed on a first come, first served basis, and is dependent on Technical Inspection lane availability. Teams with Scheduled Inspections times will be given priority over teams presenting for reinspection.

Once the team have been released from the inspection, make note of the team's progress on the central inspection record board.